SLIDE 1: TITLE & PRESENTATION DETAILS

B.Sc. Graduation Project Defense Presentation

• **Project Title:** DESIGN AND IMPLEMENTATION OF AN INTELLIGENT DOCUMENT ANALYSIS AND LEARNING ASSISTANT SYSTEM (AuraQA)

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• **Degree & Session:** Bachelor of Science (B.Sc.) Computer Science Defense Presentation (July, 2026)

SLIDE 2: SHORT ABSTRACT

Executive Summary of the Research Project

• **Problem Context:** Cloud-based LLMs present severe data privacy risks and high operational API costs when processing private academic documents.

• **Proposed Solution:** AuraQA introduces a fully offline Retrieval-Augmented Generation (RAG) system utilizing FAISS dense vector search and local LLMs.

• **Key Methodology:** Employs Prototyping SDLC, aligned 1000-char text chunking, and a deterministic rule-based query classifier (0.00s latency).

• **Empirical Results:** Overall RAG CPU latency reduced by 48.57% (71.69s to 36.87s) with a 0.40 similarity threshold out-of-scope grounding shield.

• **Multilingual Support:** Native support for Hausa, Arabic, French, Spanish, German, and English query handling.

SLIDE 3: AIM AND OBJECTIVES OF THE SYSTEM

System Goals and Functional Capabilities

- **Overall Aim:** Develop a high-speed, secure, offline Intelligent Document Analysis and Learning Assistant System using local RAG.
- **Objective 1 (Multi-format Parsing):** Support seamless text ingestion from PDF, DOCX, TXT, and CSV files with page/row tracking.
- **Objective 2 (Local RAG QA):** Provide context-grounded question answering using Ollama local models (Qwen 2.5 / Gemma 2).
- **Objective 3 (Study Mode Tools):** Generate structured summaries, multiple-choice quizzes, and interactive study flashcards offline.
- **Objective 4 (Grounding Shield & Multilingual):** Enforce a 0.40 similarity threshold to block out-of-scope hallucinations and support Hausa QA.

SLIDE 4: SYSTEM DEVELOPMENT LIFE CYCLE (SDLC) & METHODOLOGY

Prototyping Methodology & Phase Execution

• **What is the Methodology?** Prototyping SDLC — an iterative 4-phase software engineering model chosen over rigid Waterfall for continuous experimental refinement.

• **Phase 1 - Requirements & Analysis:** Defined offline data privacy requirements, multi-format ingestion needs (PDF, DOCX, CSV), and local CPU hardware latency constraints.

• **Phase 2 - Quick Prototype Design:** Built initial Next.js single-page UI, FastAPI backend services, SQLite database schemas, and FAISS vector indexing.

• **Phase 3 - Iterative Refinement & Benchmarking:** Iteratively benchmarked chunk sizes (finding 1000 chars + 150 overlap optimal) and created the 0.00s deterministic query classifier.

• **Phase 4 - Final System Evaluation & Testing:** Validated out-of-scope grounding shield accuracy (threshold 0.40), Hausa query detection, and full offline execution.

SLIDE 5: TESTING AND BENCHMARK RESULTS

Empirical Performance Evaluation & Latency Gains

- **RAG Execution Latency:** Overall RAG execution time on CPU reduced by 48.57% (from 71.69s down to 36.87s).
- **Classifier Speedup:** Deterministic query router reduced intent classification latency from 12.4s down to 0.00s.
- **Grounding Shield Accuracy:** 0.40 cosine similarity threshold effectively blocked 100% of out-of-scope hallucinated queries.
- **System Reliability:** All primary modules (Auth, Upload, QA, Summary, Quiz, Flashcards) passed automated verification.

SLIDE 6: CONCLUSION AND CONTRIBUTIONS

Summary of Accomplishments and Significance

• **Summary:** AuraQA successfully proves that high-performance, secure AI document intelligence is achievable locally on CPU hardware.

• **Data Sovereignty:** Eliminates all third-party cloud data transmission, ensuring 100% user data privacy.

• **Zero API Operational Cost:** Provides educational institutions with unlimited AI assistance at zero API usage fees.

• **Unified Learning Platform:** First system to integrate multi-lingual document QA, automated summarization, quizzes, and flashcards in one offline application.

SLIDE 7: KEY REFERENCES

Short Academic Literature Citations

• **Lewis et al. (2020):** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems (NeurIPS).

• **Johnson et al. (2019):** Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 535-547 (FAISS Framework).

• **Reimers & Gurevych (2019):** Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. Proceedings of EMNLP.

• **Vaswani et al. (2017):** Attention Is All You Need. Advances in Neural Information Processing Systems (Transformer Architecture).

SLIDE 8: RECOMMENDATIONS & FUTURE WORK

Future Extensions & Defense Panel Q&A;

- **Recommendation 1 (OCR Integration):** Incorporate Optical Character Recognition (Tesseract/PaddleOCR) for scanned paper documents.
- **Recommendation 2 (Mobile App):** Develop a lightweight cross-platform mobile application powered by ONNX runtime.
- **Recommendation 3 (Voice Assistant):** Integrate local Whisper Speech-to-Text for hands-free voice interaction.
- **Acknowledgement & Q&A;:** Thank you for your time and attention. I am now open to questions from the honorable panel.